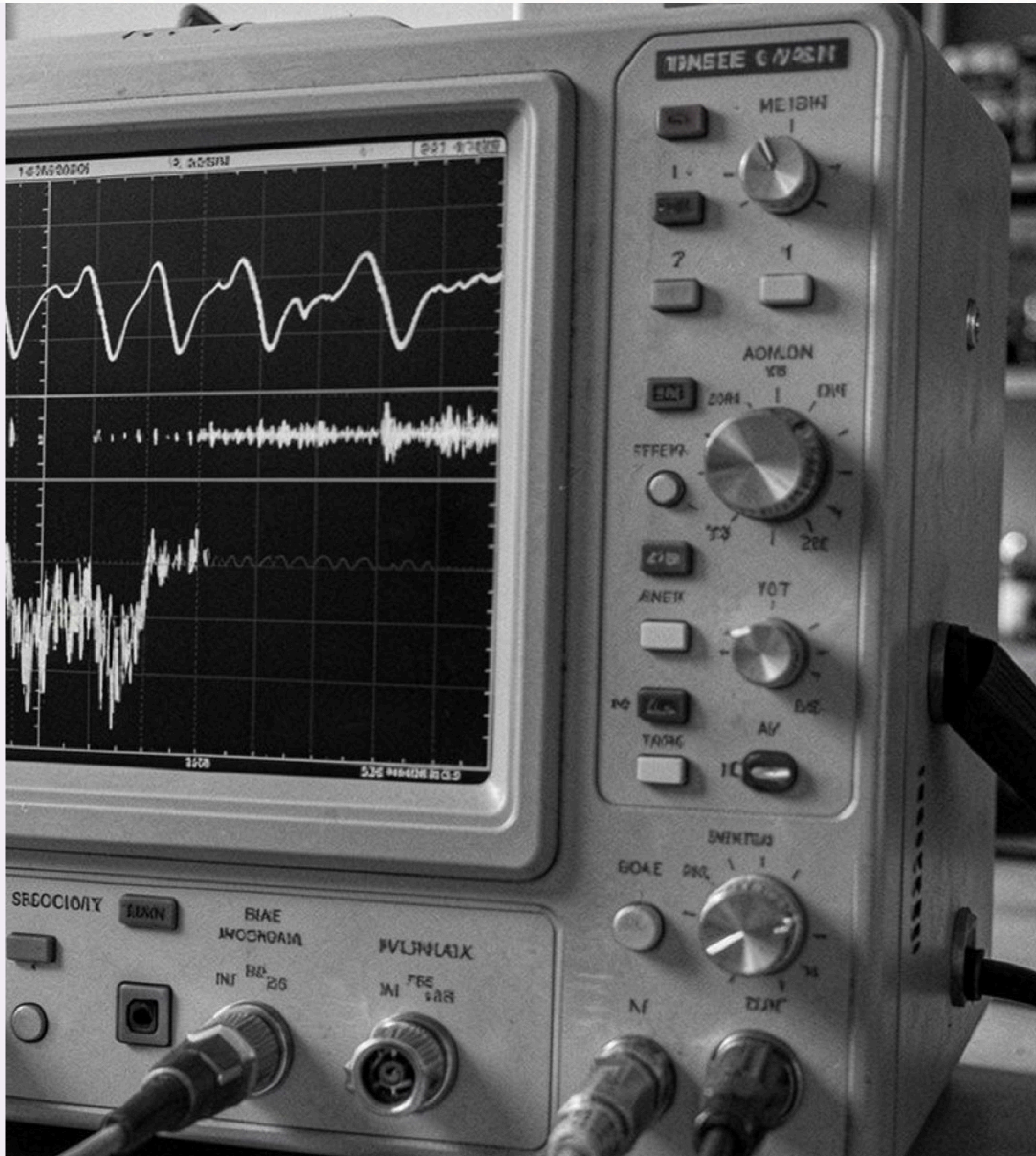
A D PATEL INSTITUTE OF TECHNOLOGY
(constituent college of CVM University)
Department of Electronics and Communication Engineering

Deterministic and Random Signals

Signals and Systems • 3rd-semester Electronics and Communication Engineering

A VISUAL INTRODUCTION TO PREDICTABILITY, UNCERTAINTY AND ENGINEERING PRACTICE

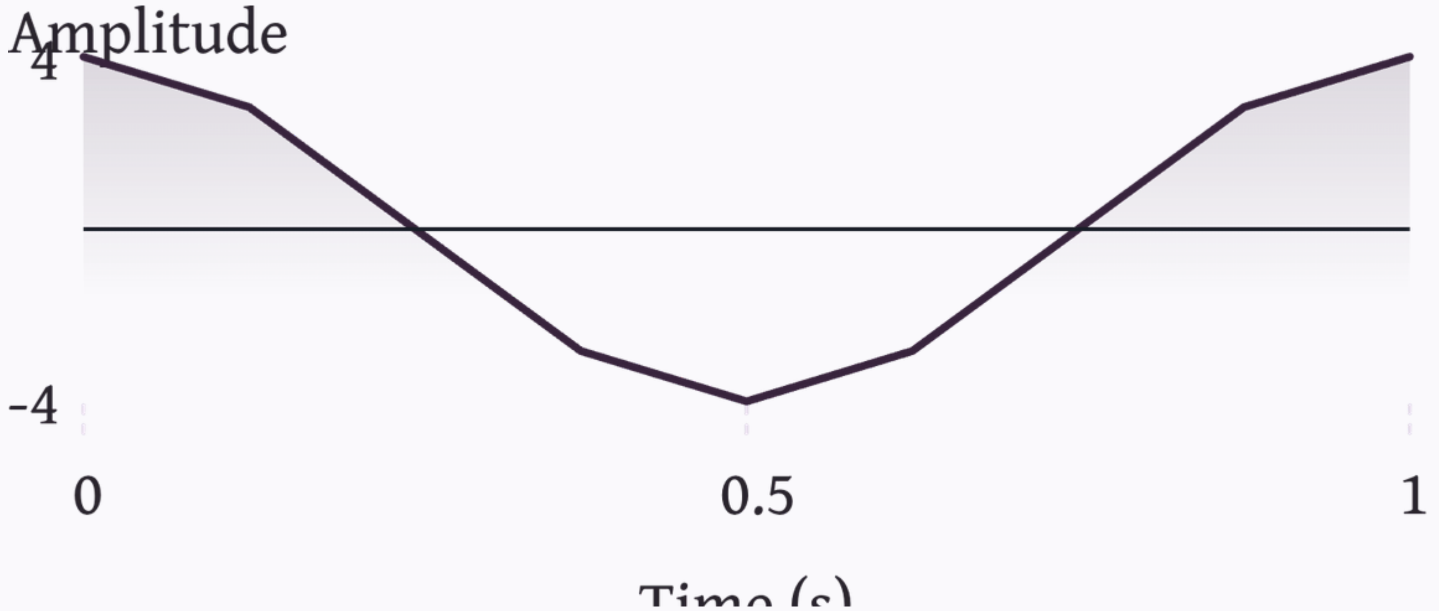


TWO DESCRIPTIONS

Two ways to describe a signal

The key difference is whether the exact future value follows from a known rule.

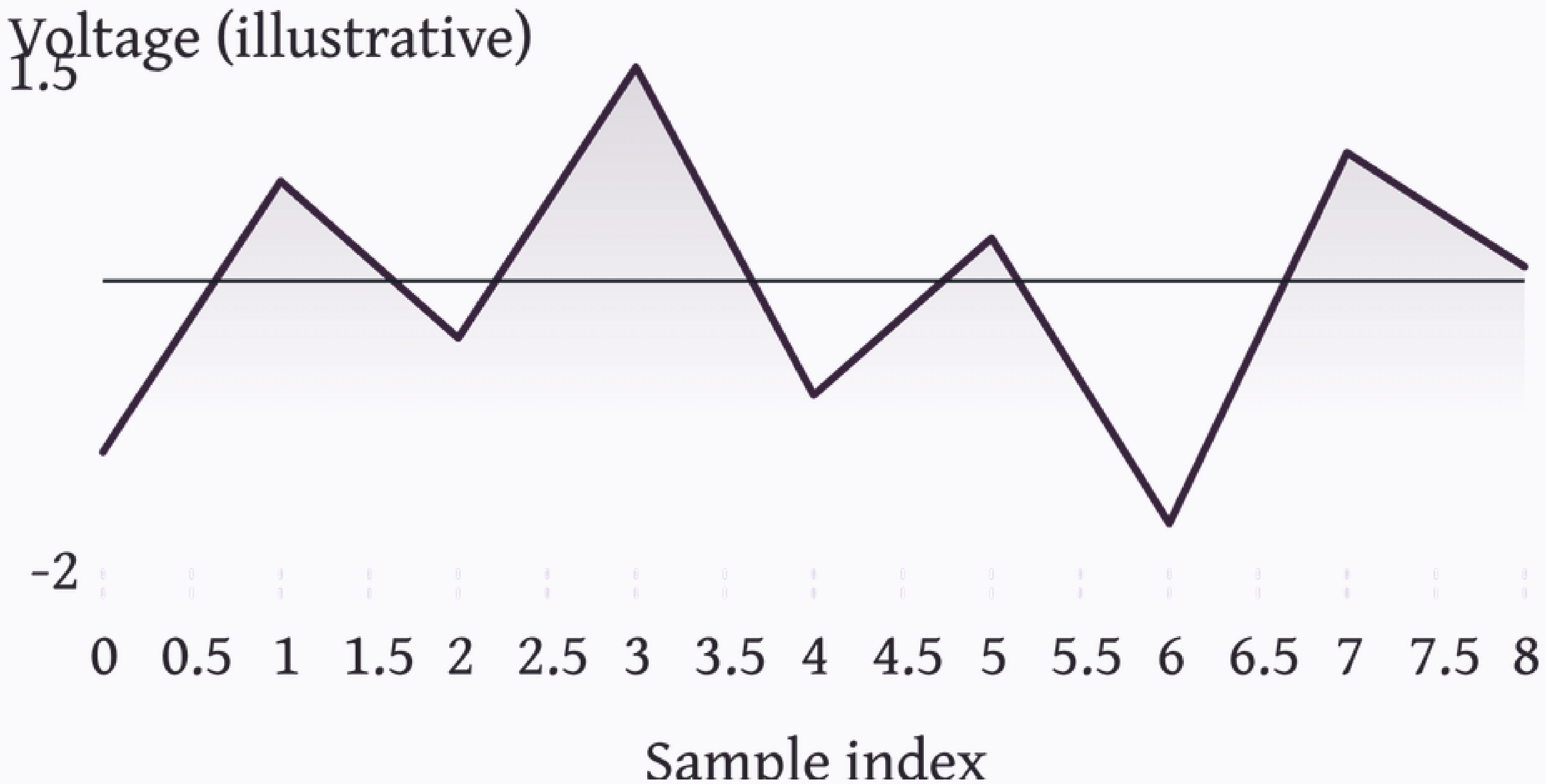
DETERMINISTIC SIGNAL



Known exactly

A mathematical rule makes the value predictable exactly at every time; amplitude, frequency, phase and waveform are known. Example: $x(t) = 4 \cos(2\pi t)$.

RANDOM SIGNAL



Known statistically

An individual realization cannot be predicted exactly in advance. Random signals are characterized by mean, variance, probability distribution, autocorrelation and power spectral density.

SOLVED EXAMPLE

Evaluate $x(t) = 4\cos(2\pi t)$

Find the signal value at $t = 0.25$ s.

01

Given

$x(t) = 4\cos(2\pi t)$

Find $x(0.25\ s)$.

02

Substitute

$x(0.25) = 4\cos(2\pi \times 0.25)$

03

Calculate phase

$2\pi \times 0.25 = \pi/2$ rad

04

Evaluate

$\cos(\pi/2) = 0$

$x(0.25) = 4 \times 0 = 0$

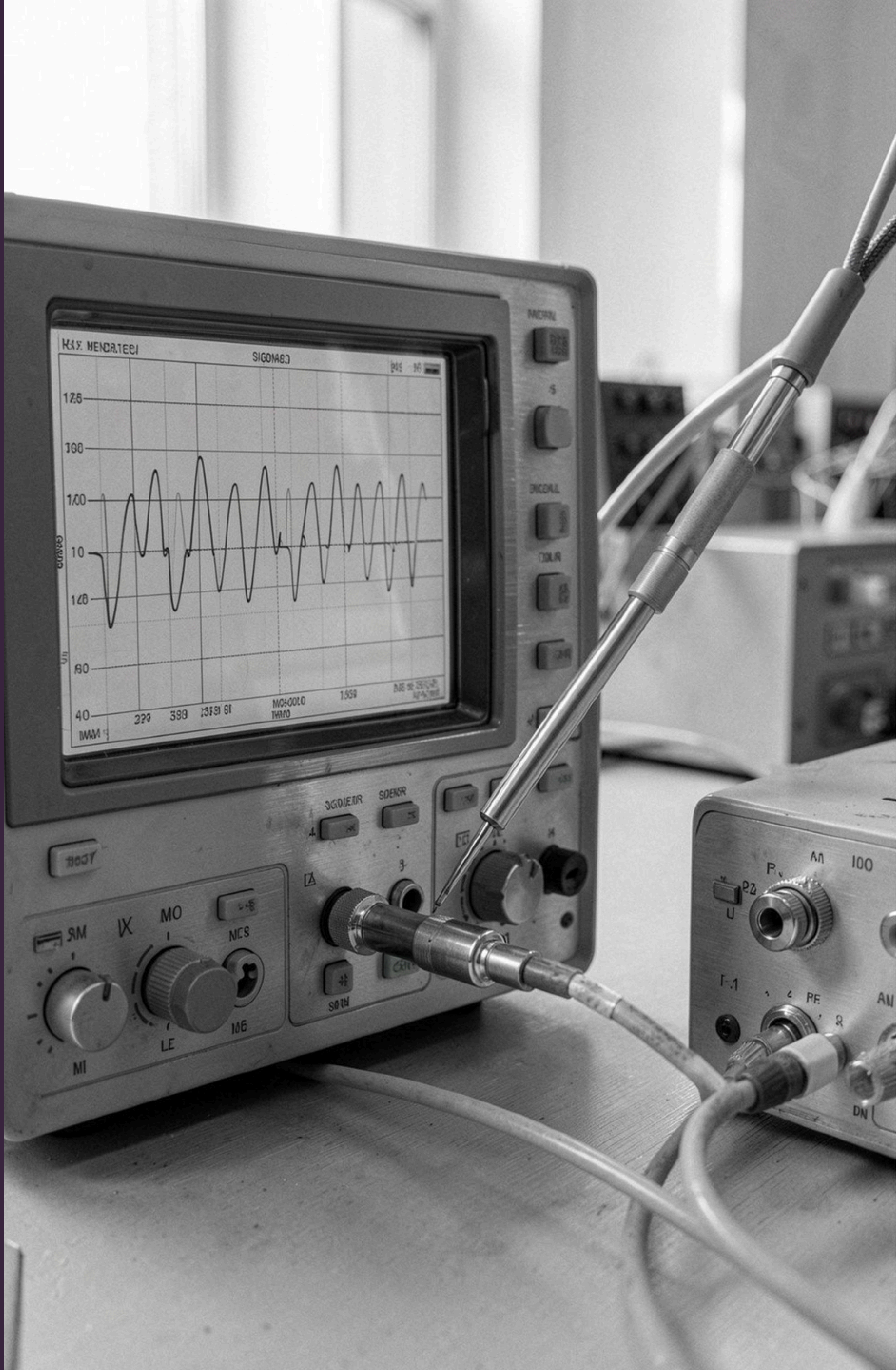
FINAL ANSWER

$x(0.25\ s) = 0$ units

The signal is deterministic because the same time input always produces the same exactly calculable output.

YOUR TURN

Can you find *x*(0.5 *s*)?



For $x(t) = 6 \sin(\pi t + \pi/6)$

Show the phase substitution clearly. Evaluate the sine term and report the final amplitude value. State why the result is predictable.

ENGINEERING PRACTICE

Where engineers use both signal types

Controlled signals test a system. Random models explain what the real channel adds.

Signal generators

Create deterministic test waveforms with controlled amplitude and frequency.

Amplifier testing

Apply known sinusoids to measure gain, bandwidth, distortion and phase response.

Channel noise

Model thermal, shot and interference effects as random disturbances.

Noise filtering

Use statistical noise models to guide filters, denoisers and SNR design.

Wireless links

Account for random fading and noise in detection, coding and reliability.

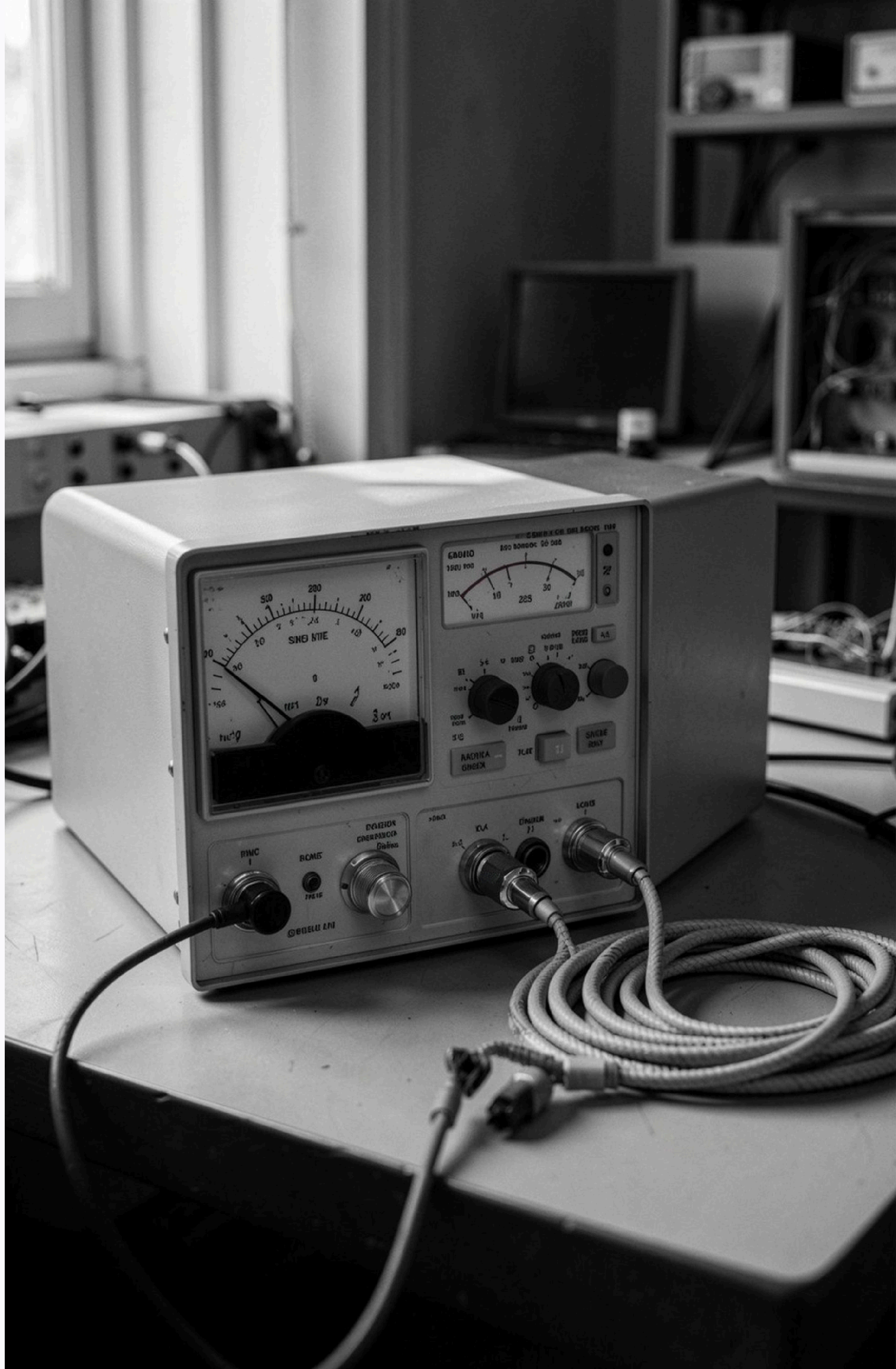
Signal processing

Use deterministic models for system response and random models for estimation and spectra.

KEY TAKEAWAYS

Predict the rule.
Model the uncertainty.

- Deterministic signals are exactly predictable from a known expression or rule.
- Random signals are described by probability and statistical measures rather than one exact future waveform.
- Both viewpoints are essential: deterministic signals build intuition for system behavior, while random signals model noise, uncertainty and real communication channels.
- In practice, engineers test with controlled signals and design for random disturbances.

THANK YOU